

* state radix or base of decimal, binary, octal and Hexadecimal number system

Radix or base

Decimal - 10	Octal - 8
Binary - 2	Hexadecimal - 16

* Binary subtraction using 1's & 2's Complement
 $(38)_{10} - (29)_{10}$

Binary of 38 = 100110 ——— ①

Binary of 29 = 11101 = 011101 ——— ②

1's complement of 011101 = 100010 ——— ③

Add ① & ③

$$\begin{array}{r} 100110 \\ + 100010 \\ \hline 1001000 \end{array}$$

For 1's complement method carry is added into ^{remaining} answer

$$\begin{array}{r} 001000 \\ + 1 \\ \hline 001001 \rightarrow (9)_{10} \leftarrow \text{Answer} \end{array}$$

1's
Complement
method

2's complement method

Take 2's complement of $(29)_{10}$

1's complement from eqn ③ $\rightarrow 100010$

$$\begin{array}{r} 2's \text{ complement of } (29)_{10} = 100010 \\ + 1 \\ \hline 100011 \end{array} \text{ ——— ④}$$

Add eqn ① & ④

$$\begin{array}{r} 100110 \\ + 100011 \\ \hline 1001001 \end{array}$$

For 2's complement method, carry is discarded

$\therefore \text{Ans} = 001001 = (9)_{10}$

* Symbol & truth tables of logic gates

Basic gates → NOT
AND
OR

universal gates
NAND
NOR

Derived
EX-OR
EX-NOR

NOT gate

Symbol

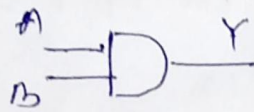


$$Y = \bar{A}$$

Truth table

A	Y
0	1
1	0

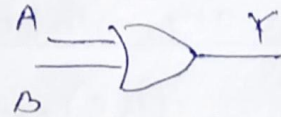
AND gate



$$Y = AB$$

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

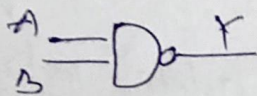
OR Gate



$$Y = A + B$$

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

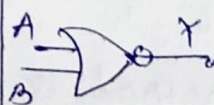
NAND gate



$$Y = \overline{AB}$$

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

NOR gate



$$Y = \overline{A + B}$$

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

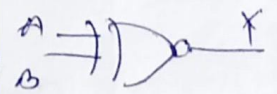
EX-OR



$$Y = A \oplus B = A\bar{B} + \bar{A}B$$

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

EX-NOR



$$Y = A \odot B = AB + \bar{A}\bar{B}$$

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

* state & prove demorgan's Theorem

→ 1st Theorem $\overline{AB} = \bar{A} + \bar{B}$

compliment of multiplication is equal to addition of each compliment of each.

2nd Theorem

$$\overline{A + B} = \bar{A} \cdot \bar{B}$$

compliment of addition is equal to multiplication of compliment of each.

Proof for demorgan's Theorem

A	B	$\bar{A}$	$\bar{B}$	$\overline{AB}$	$\bar{A} + \bar{B}$	$\overline{A+B}$	$\overline{A \cdot B}$
1	2	3	4	5	6	7	8
0	0	1	1	1	1	1	1
0	1	1	0	1	1	0	0
1	0	0	1	1	1	0	0
1	1	0	0	0	0	0	0

From column no (5) & (6) $\boxed{\overline{AB} = \bar{A} + \bar{B}}$

From column no (7) & (8) $\boxed{\overline{A+B} = \bar{A} \cdot \bar{B}}$

* state associative, commutative, distributive & inversion law

Associative law	commutative law	distributive	inversion law
$(A \cdot B) \cdot C = A \cdot (B \cdot C)$ $(A + B) + C = A + (B + C)$	$AB = BA$ $A + B = B + A$	$A(B + C) = AB + AC$	$\bar{\bar{A}} = A$

* Prove $A + AB = A$

$$\begin{aligned}
 \text{L.H.S} &= A + AB \\
 &= A(1 + B) \\
 &= A \cdot 1 \\
 &= A \\
 &= \text{R.H.S}
 \end{aligned}$$

Prove $A + \bar{A}B = A + B$

$$\begin{aligned}
 \text{L.H.S} &= A + \bar{A}B \\
 \text{put } A &= A + AB \\
 \therefore &= A + AB + \bar{A}B \\
 &= A + B(A + \bar{A}) \\
 &= A + B \cdot 1 \\
 &= A + B \\
 &= \text{R.H.S}
 \end{aligned}$$

Prove that

$$\begin{aligned}
 (A + B)(A + C) \\
 &= A + BC
 \end{aligned}$$

$$\begin{aligned}
 \text{LHS} &= (A + B)(A + C) \\
 &= AA + AC + AB + BC \\
 &= A + AC + AB + BC \\
 &= A(1 + C) + AB + BC \\
 &= A + AB + BC \\
 &= A(1 + B) + BC \\
 &= A + BC \\
 &= \text{R.H.S}
 \end{aligned}$$

Explain Universal Gate or (NAND & NOR as an universal gate with examples)

→ Universal Gate.

These gates are used to implement all other gates, hence it known as universal gate.

NAND and NOR are an universal Gates

① NAND as an universal gates.

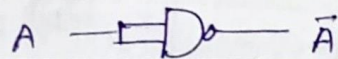
a) NOT using NAND gate

$$\text{For NAND, } Y = \overline{AB}$$

if both input are same then

$$\therefore Y = \overline{AA} = \overline{A}$$

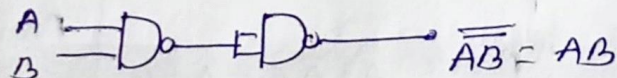
Thus, by connecting both input of NAND gate together, it form NOT gate



b) AND using NAND

$$\text{For AND, } Y = AB$$

$$\therefore Y = \overline{\overline{AB}} \quad (\because \text{Inversion law})$$



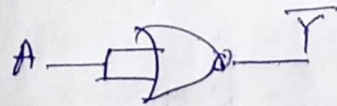
② NOR as an universal gate

a) NOT using NOR gate

$$\text{For NOR, } Y = \overline{A+B}$$

$$\text{if } A=B, \text{ then } Y = \overline{A+A}$$
$$Y = \overline{A}$$

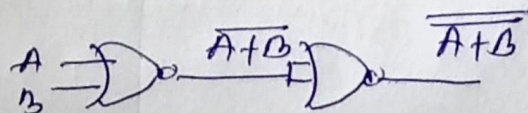
connect both input together
it form NOT gate



b) OR using NOR

$$\text{For OR, } Y = A+B$$

$$Y = \overline{\overline{A+B}}$$



c) AND using NOR

$$Y = AB$$

$$= \overline{\overline{AB}}$$

$$= \overline{A+B}$$

